Master's Encyclopedia

Contents

1 Introduction

The purpose of this document is to provide detailed explanations the most important points of the paper's provided by Dr. Protas for my masters thesis research. Each section is intended to provide the relevant definitions, derivations, and theory of the topic as it relates to my research. This document is not intended as a comprehensive review of each subject. Segments of this document are taken directly from textbooks, papers, and online sources and so is not necessarily my original work. Sections which are my original work are labelled as such.

2 3D Torus

The 3D torus is denoted by the symbol, $\Omega = \mathbb{T}^3 := (\mathbb{R}/\mathbb{Z})^3$. The domain of the torus is usually $\boldsymbol{x} \in [0,1]^3$ or $\boldsymbol{x} \in [0, 2\pi]^3$ where $\boldsymbol{x}$ is a vector. The 3D torus has periodic boundaries. The functions defined on Ω are usually periodic, but do not necessarily need to be.

Because of the periodic boundary, each pair of opposite faces on Ω are actually the same face. Therefore, the value of a periodic function is the same on opposite faces. However, the normal vector in the surface integral is, by convention, outward-pointing. Therefore, the surface integral over one face will be exactly cancelled by the integral over the opposite face. This means the surface integral of a periodic function on Ω is zero.

The resolution of a simulation describes the number of points in the grid. It is denoted by the symbol N^3 where N is the number of points along each axis. Typically the simulations will use a resolution of 128^3, 256^3, 512^3, or 1024^3. The simulations for my master's thesis will use uniform grids. However, some simulations utilize adaptive grids which increase the local resolution near points of interest.

3 Smooth and Analytic Functions

The smoothness of a function is a property measured by the number of continuous derivatives it has over its domain. The smoothness of a function is denoted by the symbol, C^k where the function has a k-th order continuous derivative. The function of class C^∞ is an infinitely differentiable function.

An analytic function is a function that is differentiable at every point in a region, and it can be represented by a convergent power series. There are both real and complex analytic functions. Both types are infinitely differentiable and so are in C^∞. However, not all C^∞ functions are analytic. For example,

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

This function is smooth as it has continuous derivatives of all orders at every real number x. However, the function is not analytic because there is no convergent power series at $x = 0$.

4 Fourier Analysis

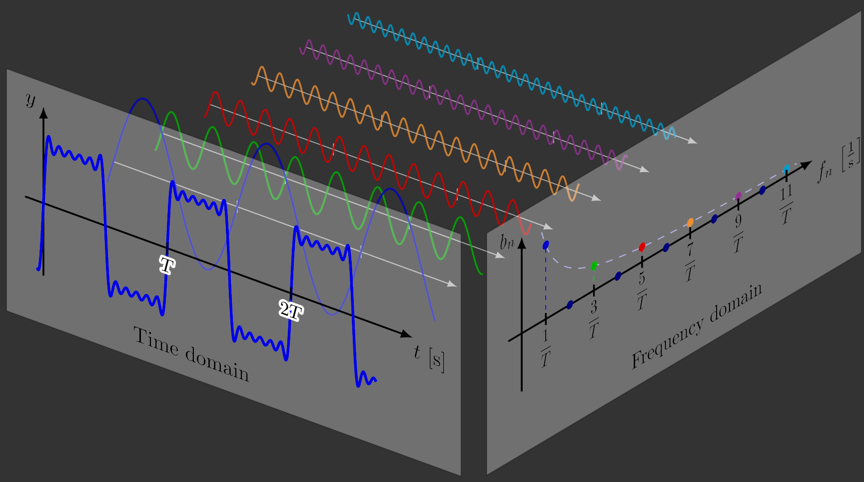
In mathematics, Fourier analysis is the study of the way general functions may be represented or approximated by sums of simpler trigonometric functions. We make use of several different components of Fourier Analysis.

4.1 Fourier Transform

The Fourier Transform (FT) is a mathematical tool that decomposes a function into its constituent frequencies. This transforms the function from physical space to its frequency domain or Fourier Space. The Fourier transform of a function $u(x)$, $x \in \mathbb{R}$ is the function $\hat{u}(k)$ defined by,

$$\hat{u}(k) = \int_{-\infty}^{\infty} e^{-ikx} u(x) dx, \qquad k \in \mathbb{R}.$$

The number $\hat{u}(k)$ can be interpreted as the amplitude density of u at wavenumber k. Below is a visualization of the Fourier transform of a function.



4.2 Inverse Fourier Transform

Conversely, we can reconstruct u from $\hat{u}$ by the inverse Fourier transform,

$$u(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \hat{u}(k) dk, \qquad x \in \mathbb{R}.$$

The Fourier transform and inverse Fourier transform are defined over continuous unbounded space (x) and continuous unbounded wavenumbers (k). The L^2-norms of u and $\hat{u}$ are related by Parseval's equality,

$$\|\hat{u}\| = \sqrt{2\pi} \|u\|$$

4.3 Semi-discrete Fourier Transform

In numerical simulations we utilize discrete space, not continuous space. Therefore, we must consider $x \in h\mathbb{Z}$ rather than $x \in R$. Precise analogues of the Fourier transform and its inverse exist for this case. However, because the spatial domain is discrete, the wave number k will no longer range over all of $\mathbb{R}$. Rather, the wavenumber domain is a bounded interval of length $2\pi/h$. One suitable case for this the range $x \in [-\pi/h, \pi/h]$. For a function v defined on $h\mathbb{Z}$ with value v_j at x_j, the semi-discrete Fourier transform is defined by,

$$\hat{v}(k) = h \sum_{-\infty}^{\infty} e^{-ikx_j} v_j, \qquad k \in [-\pi/h, \pi/h].$$

4.4 Inverse Semi-discrete Fourier Transform

The inverse semi-discrete Fourier transform is,

$$v_j = \frac{1}{2\pi} \int_{-\pi/h}^{\pi/h} e^{ikx_j} \hat{v}(k) dk, \qquad j \in \mathbb{Z}.$$

The Semi-discrete Fourier Transform and Semi-discrete Inverse Fourier Transform are defined over unbounded discrete space (x) and continuous bounded wave numbers (k).

4.5 Discrete Fourier Transform

Numerical simulations not only utilize discrete space, but finite discrete space. Therefore we must consider $x \in h\mathbb{Z}$ for a finite number of terms. For this case we use the Discrete Fourier Transform,

$$\hat{v}(k) = \sum_{j=0}^{N-1} v_j e^{-2\pi ikj/N}, \qquad j \in \mathbb{Z}$$

4.6 Inverse Discrete Fourier Transform

The inverse discrete Fourier transform is,

$$v_j = \frac{1}{N} \sum_{k=0}^{N-1} \hat{v}(k) e^{2\pi ikj/N}, \qquad j \in \mathbb{Z}$$

The Discrete Fourier Transform and Discrete Inverse Fourier Transform are defined over bounded discrete space (x) and bounded discrete wave numbers (k).

4.7 Fourier Series

A Fourier series is an expansion of a periodic function $f(x)$ into a sum of trigonometric functions. The formula for a Fourier series is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L)]$$

where a_n and b_n are the Fourier Coefficients given by the formulas,

$$a_0 = \frac{1}{L} \int f(x) dx$$

$$a_n = \frac{1}{L} \int_{2L} f(x) \cos(n\pi x/L) dx$$

$$b_n = \frac{1}{L} \int_{2L} f(x) \sin(n\pi x/L) dx$$

4.8 Fourier Expansion

We must also consider the case of a function defined on a periodic domain $x \in \Omega$. A function in this domain can be expressed with the Fourier expansion,

$$u(x) = \sum_{k \in \mathbb{Z}^3} \hat{u}_k e^{ikx}$$

where the coefficients $\hat{u}_k$ are complex numbers satisfying the condition

$$\hat{u}_k = \overline{\hat{u}_{-k}} \qquad \text{for all} \qquad k \in \mathbb{Z}^3$$

imposed to ensure that u is real. If needed, the Fourier coefficients $\hat{u}_k$ can be computed explicitly via the integral,

$$\hat{u}_k = \frac{1}{(2\pi)^3} \int_\Omega e^{-ikx} u(x) dx.$$

The Fourier coefficients represent the amplitude and phase at each frequency of the series. From the Payley-Wiener theorem, if $u(x)$ is real analytic, then its Fourier coefficients decay exponentially and conversely, if a Fourier expansion has exponentially decreasing coefficients then it is real analytic.

4.9 Fast Fourier Transform

The Fast Fourier Transform is an algorithm for computing the discrete Fourier transform. The most common FFT algorithm is the Cooley-Tukey algorithm.

4.10 Inverse Fast Fourier Transform

The Inverse Fast Fourier Transform is an algorithm for computing the inverse discrete Fourier transform.

4.11 Fourier Filter

The Fourier filter is a type of filtering function that is based on manipulation of specific frequency components of a signal. It works by taking the Fourier transform of the signal, then attenuating or amplifying specific frequencies, and finally inverse transforming the result.

4.12 Randomization in Fourier Space

For my thesis we will use random perturbations and initial conditions. However, a randomly generated function is not likely to be divergence free, analytic, or have a mean of zero. In order to achieve this we make use of the Payley-Wiener theorem described above. We created a Fourier expansion with exponentially decreasing coefficients to ensure it is analytic and randomly change the phase of the Fourier coefficients to create our random function. Then we apply Hodge Projection to ensure it is divergence free. Given a vector v such that $\nabla \cdot v$, we define the divergence free vector u,

$$u = Pv = v - \nabla \Delta^{-1} \nabla \cdot v$$

Aliasing happens when high-frequency components are mistaken as lower frequencies due to insufficient sampling. We apply dealiasing in Fourier space. Finally we apply the Inverse Fast Fourier Transform to convert back to physical space.

5 Initial Conditions

There are a variety of initial conditions that can be used in our research. The initial condition is denoted by the symbol, η. There are several key properties common to these initial conditions.

The first is that each initial condition is periodic on Ω. We also assume, without loss of generality, that

$$\int_\Omega \eta dx = 0$$

WHAT IS IMPORTANT ABOUT THIS. This property remains true for all time t. The initial conditions must be divergence free,

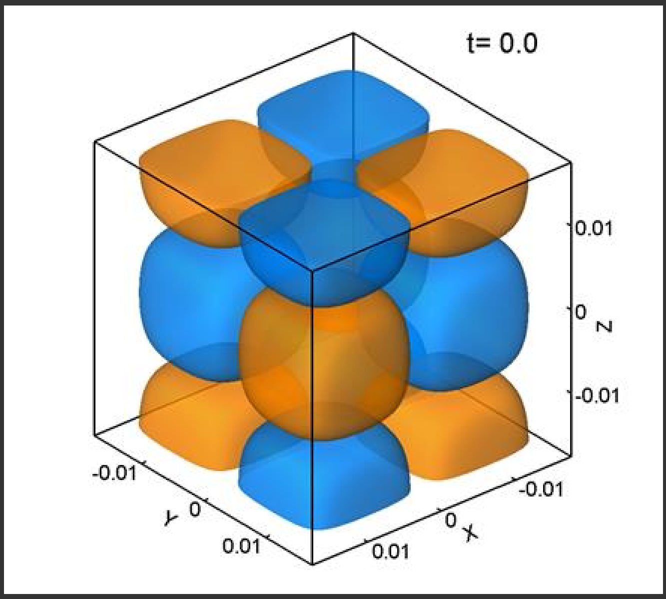
$$\nabla \cdot \eta = 0$$

5.1 Taylor-Green Vortex

The Taylor–Green vortex has been widely used as a candidate for potential blow up in Euler flows, however,it is still an open question whether this initial condition can indeed lead to a singularity formation in a finite time. The formula for the Taylor-Green vortex is,

$$\eta := \begin{bmatrix} \sin(2\pi)\cos(2\pi y)\cos(2\pi z) \\ -\cos(2\pi x)\sin(2\pi y)\cos(2\pi z) \\ 0 \end{bmatrix}$$

Below is a visualization of the z-component vorticity of the Taylor-Green Vortex.



5.2 Kerr

The Kerr initial condition represents two perturbed anti-parallel vortex tubes located symmetrically with respect to the xy-plane such that $\omega(x, y, z) = -\omega(x, y, -z)$. It is given by the formula,

$$\eta := \nabla \times (|D|^{-2}\omega$$

where

$$\omega = 8G\omega(r)[\omega_1, \omega_2, \omega_3]^T$$

where G is the Fourier filter defined by

$$\hat{G}_j = \exp(-0.05(j_1^4 + j_2^4 + j_3^4))$$

and

$$\omega(r) = \begin{cases} \exp[(-r^2/(1-r^2)) + r^4(1+r^2+r^4)] & r < 1, \\ 0 & r \geq 1. \end{cases}$$

where

$$r = \frac{1}{R}\sqrt{[x - x(s)]^2 + [z - z(s)]^2}$$

$$x(s) = \delta_x \cos(\pi s/L_x)$$

$$z(s) = z_0$$

$$s(y) = y_2 + L_y \delta_{y1} \sin(\pi y_2/L_y)$$

$$y_2(y) = 4\pi y + L_y \delta_{y2} \sin(4\pi y)/L_y)$$

$$\omega_1 = \frac{-\pi\delta_x}{L_x}[1 + \pi\delta_{y2}\cos(\pi(4\pi y)/L_y)]] * [1 + \pi\delta_{y1}\cos(\pi y_2/L_y)]\sin(\pi s/L_x)$$

$$\omega_2 = 1$$

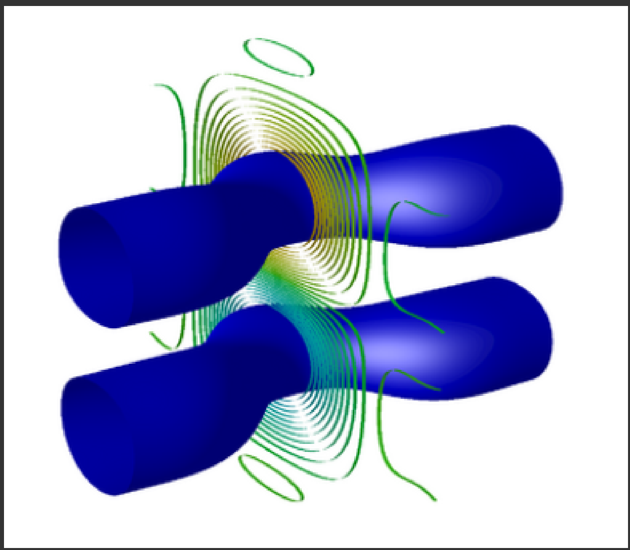
$$\omega_3 = 0$$

We choose the parameters,

$$\delta_{y1} = 0.5, \delta_{y2} = 0.4, \delta_x = -1.6, z_0 = 1.57$$

$$R = 0.75, L_x = L_y = 4\pi, L_z = 2\pi$$

A visualization of the vorticity of the Kerr initial condition is shown below. The tube is the isosurface at 60% of the maximum vorticity.



5.3 Hou

The Hou axisymmetric initial condition is given by the equation,

$$\eta := G u_\theta e_\theta$$

where e_θ is the unit vector in the azimuthal direction of the cylindrical coordinate system and u_θ is the angular velocity component given by,

$$u_\theta = \begin{cases} r \exp(-r^2/(1-r^2)) \frac{12000(1-r^2)^{18} \sin(2\pi z)}{1+12.5\sin^2(\pi z)} & r < 1, \\ 0 & r \geq 1. \end{cases}$$

where

$$r = \frac{\sqrt{x^2+y^2}}{0.9}$$

A visualization of the Hou initial condition is shown below. FIND VISUALIZATION.

5.4 Random

The random initial condition is also described in the Randomization in Fourier Space section and is defined by the equation,

$$\eta := P_L v_{\text{rand}} := v_{\text{rand}} - \nabla\Delta^{-1}(\nabla \cdot v_{\text{rand}})$$

where P_L is the Leray projector with the property that for any $\omega \in H^1(\mathbb{T}^3)$, $\nabla \cdot (P_L \omega) = 0$, and v_{rand} is given by,

$$v_{\text{rand}} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} \sum_{|j_1|+|j_2|+|j_3| \leq N_0} e^{-|j|/(4\pi)} e^{i2\pi\theta_1(j_1)} e^{i2\pi j \cdot x} + c.c. \\ \sum_{|j_1|+|j_2|+|j_3| \leq N_0} e^{-|j|/(4\pi)} e^{i2\pi\theta_2(j_2)} e^{i2\pi j \cdot x} + c.c. \\ \sum_{|j_1|+|j_2|+|j_3| \leq N_0} e^{-|j|/(4\pi)} e^{i2\pi\theta_3(j_3)} e^{i2\pi j \cdot x} + c.c. \end{bmatrix}$$

where θ_1, θ_2, and θ_3 are N_0-dimensional random variables with uniform distributions on $[0,1]^{N_0}$. In practice, we use $N_0 = 64$.

6 Supremum and Infimum

The supremum, denoted as $\sup f(x)$, represents the smallest upper bound of the values attained by the function over a given domain. It is the largest value that the function can achieve. The infimum, denoted as $\inf f(x)$, represents the greatest lower bound of a set, meaning the largest number that is less than or equal to all elements in the set.

The essential supremum, $\operatorname{ess\,sup} f(x)$, is the smallest value that is greater than or equal to the function values everywhere while ignoring what the function does at a set of points of measure zero. The essential infimum, $\operatorname{ess\,inf} f(x)$, is the largest value that is greater than or equal to the function values everywhere while ignoring what the function does at a set of points of measure zero.

The limit superior, denoted $\limsup f(x)$, is the supremum of a functions limit. Therefore it is the smallest upper bound of the values attained by the function over a given domain as it approaches the limit argument. The limit infimum, denoted $\liminf f(x)$, is the infimum of a functions limit. Therefore it is the largest lower bound of the values attained by the function over a given domain as it approaches the limit argument.

7 Functional Analysis

Functions can be thought of as points in an infinite dimensional space of vectors. We must first define some set operations. The product of two sets is the set of pairs,

$$X \times Y := \{(x, y) : x \in X, y \in Y\}$$

A function $f : X \to Y, x \mapsto f(x)$, assigns, for every input $x \in X$, a unique output element $f(x) \in Y$. X is called the domain of f and Y its codomain.

7.1 Metric Spaces

A metric space is a set X with a metric d,

$$d : X \times X \to [0, \infty)$$

with the conditions that, given $x, y, z \in X$,

$$d(x, y) = 0 \Leftrightarrow x = y$$

$$d(x, y) = d(y, x)$$

$$d(x, y) \leq d(x, z) + d(z, y)$$

An open ball around x is the set,

$$B_\epsilon(x) := \{y \in X | d(x, y) < \epsilon\}$$

where $\epsilon > 0$. A set $A \subset X$ is called open if for each $x \in A$ there is an open ball with $B_\epsilon(x) \subset A$. A boundary point is a point $x \in X$ is a boundary point for A if for all $\epsilon > 0$,

$$B_\epsilon(x) \cap A \neq \emptyset$$

and

$$B_\epsilon(x) \cap A^c \neq \emptyset$$

The boundary of A is the set that contains all boundary points for A and is denoted as ∂A. A set A is closed if A^c is open. The closure of a is $\overline{A} = A \cup \partial A$. Let D be a bounded open subset of $\mathbb{R}^n$, and $\overline{D}$ be a bounded closed subset of $\mathbb{R}^n$.

7.2 Complete Space

A sequence $x_1, x_2, x_3, ... \in X$, where X is a metric space, is called Cauchy if for every positive real number r there is a positive integer N such that for all positive integers $m, n > N$, $d(x_m, x_n) < r$. A metric space is complete if every Cauchy sequence in X has a limit that is also in X.

7.3 Normed Space

A map $\| \cdot \| : X \to [0, \infty)$ is called a norm if,

$$\|x\| = 0 \Leftrightarrow x = 0$$

$$\|\lambda x\| = \lambda | \|x\|$$

$$\|x + y\| \leq \|x\| + \|y\|$$

A vector space with a norm is called a normed space. If $\| \cdot \|$ is the norm of the vector space X, then

$$d_{\|\cdot\|}(x, y) := \|x - y\|$$

defines a metric for the set X. Therefore, a normed space is a type of metric space. A non-negative function that satisfies the first two axioms is termed a seminorm.

7.4 Function Spaces

A function space is a set of functions between two fixed sets. For example, the space of bounded functions from set X to a metric space Y is itself a metric space, with distance defined by

$$d(f,g) := \sup_{x\in X} d(f(x), g(x))$$

which is complete when Y is complete.

A vector space V over a field $\mathbb{F}$ is a set on which are defined an operation of vector addition, $+ : V \times V \to V$, satisfying associativity, commutativity, zero and inverse axioms, and an operation of scalar multiplication, $\mathbb{F} \times V \to V$, that satisfies the respective distributive laws.

7.5 Banach Spaces

A Banach space is a complete normed metric space. The space $C^0(D)$ is the space of continuous functions on D. The space $C^0(\overline{D})$ of continuous functions on $\overline{D}$ with norm,

$$||u||_{C^0(\overline{D})} := \sup_{x\in\overline{D}} |u(x)|$$

is a Banach space. The space of k-times continuously differential functions on $\overline{D}$ is a Banach space when equipped with the norm,

$$||u||_{C^k(\overline{D})} := \sum_{|\alpha|\leq k} ||\partial^\alpha u||_{C^k(\overline{D})}$$

When X is a Banach space we denote the space of all continuous functions from [0,T] into X by $C([0,t];X)$, equipped with the norm,

$$\|u\|_{C([0,t];X)} := \sup_{t\in[0,T]} \|u(t)\|_X$$

7.6 Lebesgue Spaces

Lebesgue spaces are function spaces defined using a natural generalization of the p-norm for finite-dimensional vector spaces. They are a type of Banach Space. By $L^p(D)$, where $1 \leq p < \infty$, we denote the standard Lebesgue space of measurable p-integrable functions with the norm,

$$\|u\|_{L^p} := \left(\int_D |u(x)|^p dx\right)^{1/p}$$

The space $L^\infty(D)$ of essentially bounded functions on D is equipped with the standard norm

$$\|u\|_{L^\infty} := \text{ess sup}|u|$$

7.7 Sobolev Spaces

A Sobolev space is a vector space of functions equipped with a norm that is a combination of L^p-norms of the function together with its derivatives up to a given order.

Let $1 \leq p < \infty$. The space $W^{1,p}(D)$ consists of all those functions $u \in L^p(D)$ all of whose first weak derivatives $\partial_i u$ exist and are in $L^p(D)$. The norm in $W^{1,p}(D)$ is given by,

$$\|u\|_{W^{1,p}} := \left(\int_D |u(x)|^p + \sum_{i=1}^{n} \int_D |\partial_i u(x)|^p dx \right)^{1/p}$$

The space $W^{k,p}(D)$ is the space of functions $u \in W^{k-1,p}(D)$ all of whose first weak derivatives also belong to $W^{k-1,p}(D)$. The standard norm in $W^{k,p}(D)$ is given by

$$\|u\|^p_{W^{k,p}} := \sum_{|\alpha| \leq k} \|\partial \alpha u\|^p_{L^p}.$$

On Ω we can make use of the Fourier expansion of u. The L^2-norm is,

$$\|u\|^2 = (2\pi)^3 \sum_k |\hat{u}_k|^2$$

We will often use the L^2-based Sobolev spaces $W^{k,2}$. We adopt the notation,

$$H^k(D) := W^{k,2}(D)$$

$$H^k_0(D) := W^{k,2}_0(D)$$

We can use the gradient of u to obtain,

$$\|\nabla u\|^2 = (2\pi)^3 \sum_{k \in \mathbb{Z}^3} |k|^2 |\hat{u}_k|^2$$

For $s \geq 0$ the Sobolev space $H^s(\Omega)$ consists of all the functions for which the norm,

$$\|u\|^2_{H^s} := (2\pi)^3 \sum_{k \in \mathbb{Z}^3} (1 + |k|^{2s}) |\hat{u}_k|^2$$

is finite. For any $u \in H^s(\Omega)$,

$$\|u\|^2_{\dot{H}^s} := (2\pi)^3 \sum_{k \in \mathbb{Z}^3} |k|^{2s} |\hat{u}_k|^2$$

7.8 Hilbert Spaces

A Hilbert space is a vector space equipped with an inner product operation. A Hilbert space is a type of Banach space where the norm is derived from an inner product. The spaces $H^k(D)$ and $H^k_0(D)$ are Hilbert spaces with the inner product,

$$< u, v >_{H^k} := \sum_{0 \leq |\alpha| \leq k} < \partial^\alpha u, \partial^\alpha v$$

, and the corresponding norm,

$$\|u\|^2_{H^k} = < u, u >_{H^k} = \sum_{0 \leq |\alpha| \leq k} \|\partial^\alpha u\|^2.$$

The space Lebesgue space $L^2(D)$ is a Hilbert space when equipped with the inner product,

$$< u, v > := \int_D u(x) \cdot v(x) dx$$

7.9 Gevrey Subspace

The Gevrey class we utilize in the optimization problem is defined as,

$$G^\sigma := \{v \in H^s(\Omega) : \|v\|_{G^\sigma} :=< v, v >^{1/2}_{G^\sigma} < \infty\}, s \geq 3, \sigma > 0$$

with the inner product

$$\forall v, u \in G^\sigma, < v, u >_{G^\sigma} := \sum_{j \in \mathbb{Z}^3} (1 + |2\pi j|^2)^s e^{4\pi\sigma|j|} \hat{v}_j \cdot \bar{\hat{u}}_j)$$

$$= \int_\Omega (1 + |D|^2)^s e^{2\sigma|D|} v \cdot u dx$$

where overbar denotes complex conjugation, where as the operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[|\hat{D}|v\right]_j := 2\pi|j|\hat{v}_j$$

$$\left[e^{\sigma|\hat{D}|}v\right]_j := e^{2\pi\sigma|j|}\hat{v}_j.$$

7.10 Manifolds

8 Strong vs Weak Solutions

9 Time Stepping

9.1 leapfrog bk

9.2 ssprk3 bk

Strong stability preserving 3rd order Runge-Kutta. This method is called strong stability preserving because it preserves the strong stability properties of first-order Euler time stepping method.

9.3 rk4 bk

The Fourth Order Runge Kutta Method is a numerical method used to approximate solutions to ODEs. It's a highly accurate and widely used method that provides a step-by-step approach to estimating the solution's next value based on an average of slopes at different points within a small interval.

K1 is the slope at the beginning of the time step.

K2 is an estimate of the slope at the half time step using K1.

K3 is another estimate of the slope at the half time step using K2.

K4 is an estimate of the slope at the end of the time step.

The RK4 algorithm works by using the four approximations of the slope. It uses these approximations to estimate the slope with a weighted average at some time t. It then uses this estimate to update the PDE solution value.

9.4 rk5 bk

The Fifth Order Runge Kutta Method is similar to RK4 but uses six approximations of the slope and a different weighted average.

10 The Adjoint Method

10.1 The Gateaux Directional Differential

The equation for the Gateaux differential is,

$$\Phi'_{\alpha,T}(\eta;\eta') := \lim_{\epsilon\to 0}[\Phi_{\alpha,T}(\eta+\epsilon\eta') - \Phi_{\alpha,T}(\eta)].$$

The objective functional is defined as,

$$\Phi_{\alpha,T}(\eta) := \|\nabla u_\alpha(T;\eta)\|^2_{L^2} = \|u_\alpha(T;\eta)\|^2_{\dot{H}^1}.$$

Substituting the second definition into the Gateaux differential,

$$\Phi'_{\alpha,T}(\eta;\eta') = \lim_{\epsilon\to 0}[\|u_\alpha(T;\eta+\epsilon\eta')\|^2_{\dot{H}^1} - \|u_\alpha(T;\eta)\|^2_{\dot{H}^1}].$$

Substituting in the equation for the $\dot{H}^1$ seminorm,

$$\Phi'_{\alpha,T}(\eta;\eta') = \lim_{\epsilon\to 0}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{u}_k+\epsilon\hat{u}'_k|^2 - (2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{u}_k|^2].$$

This simplifies to,

$$\Phi'_{\alpha,T}(\eta;\eta') = \lim_{\epsilon\to 0}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2 2\epsilon|\hat{u}_k\hat{u}'_k| + (2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2\epsilon^2|\hat{u}'_k|^2].$$

Taking the limit,

$$\Phi'_{\alpha,T}(\eta;\eta') = 2(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{u}_k\hat{u}'_k|.$$

This can be expressed in the integral form,

$$\Phi'_{\alpha,T}(\eta;\eta') = 2\int_{\mathbb{T}^3}\sum_{k\in\mathbb{Z}^3}|k|^2\hat{u}_k\hat{u}'_k e^{2ikx}dx.$$

Which is equivalent to the inner product,

$$\Phi'_{\alpha,T}(\eta;\eta') = 2 < \sum_{k\in\mathbb{Z}^3}|k|\hat{u}_k e^{ikx}, \sum_{k\in\mathbb{Z}^3}|k|\hat{u}'_k e^{ikx} >$$

This gives us the Gateaux differential in the form of the $\dot{H}^1$ inner product,

$$\Phi'_{\alpha,T}(\eta;\eta') = 2 < u_\alpha(\cdot,T), u'_\alpha(\cdot,T) >_{\dot{H}^1}$$

11 Navier-Stokes Equations

The Navier Stokes Equations are defined as

$$\partial_t\mathbf{u} + \mathbf{u}\cdot\nabla\mathbf{u} + \nabla p - \mu\Delta\mathbf{u} = 0 \ \textbf{in} \ \Omega\times(0,T]$$

$$\nabla\cdot\mathbf{u} = 0 \text{ in } \Omega\times[0,T]$$

$$\mathbf{u}(0) = \mathbf{u}_0$$

where $\mathbf{u}:[0,T]\times\Omega\to\mathbb{R}^3$ and $p:[0,T]\times\Omega\to\mathbb{R}$ are the velocity and pressure fields. $\mu>0$ is the coefficient of kinematic viscosity, $\mathbf{u}_0$ is a divergence-free initial condition whereas the fluid density ρ is assumed constant and equal to 1.

12 Navier-Stokes-Voigt Equations

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13 Euler's Equations

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The Euler equations are the inviscid incompressible from of the Navier-stokes equations. They are defined by the equations

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p = 0$$

$$\nabla \cdot \mathbf{u} = 0$$

$$\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x}).$$

14 Euler-Voigt Equations

THIS SECTION IS COPIED FROM MY THESIS AND IS MY ORIGINAL WORK.

14.1 α-Energy

An $\alpha-$energy equality was proved to hold for solutions for the Euler-Voigt equations for all $t \in \mathbb{R}$,

$$E_\alpha(t) := \|\mathbf{u}^\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}^\alpha(t)\|_{L^2}^2 = E_\alpha(0).$$

14.2 Linearization

15 Burgers Equation

15.1 Viscid

15.2 Inviscid

16 Blow-up Criteria

There are several different criteria that are used to determine whether a flow has formed a singularity. These criteria are described below.

16.1 Beale-Kato-Majda Criterion

The Beale-Kato-Majda (BKM) criterion states that a smooth solution u of the Euler system develops a singularity at $t = T^*$ if and only if

$$\lim_{t \to T^*} \int_0^t \|\omega(\tau)\|_{L^\infty} d\tau = \infty$$

where $\omega := \nabla \times u$ is the vorticity field of the flow.

16.2 Vorticity Criterion

16.3 Voigt-type Criterion

16.4 Enstrophy

In fluid dynamics, enstrophy is a measure of the strength of vorticity, essentially the squared magnitude of vorticity. Enstrophy is defined by the equation

$$\varepsilon(u(t)) := \frac{1}{2}\int_{\Omega} |\omega(t,x)|^2 dx = \frac{1}{2}\|\nabla u(t)\|_{L^2(\Omega)}$$

where ω is again the vorticity.

17 Brent's Minimization Method

Given a function f that depends on one or more independent variables find the value of those variables where f takes on a minimum value. Use this variables calculate the minimum value of f. In general, finding a global minimum of a function is very difficult. It is made more difficult by the fact that we do not have the derivative of the function. To solve this problem we use Brent's method.

Before describing Brent's Method, we must first bracket a minimum. A minimum is known to be bracketed only when there is a triplet of points, $a < b < c$ or $c < b < a$, such that $f(b)$ is less than both $f(a)$ and $f(c)$. In this case, we know that the function has a minimum in the interval (a, c). The initial bracketing of a minimum is done by METHOD WE USED.

We must also describe the Golden Search method as it is a tool used by Brent's Method. The general procedure is to choose a new point x, either between a and b or between b and c. Suppose we make the latter choice. Then we evaluate $f(x)$. If $f(b) < f(x)$, then the new bracketing triplet is (a, b, x). If $f(b) > f(x)$, then the new triplet is (b, x, c). In all cases, the middle point of the new triplet is the abscissa whose ordinate is the best minimum achieved so far. We continue the process until the distance between the two outer points of the triplet is below some chosen tolerance. The question remains, how do we choose the new point x, given (a, b, c). The answer is to choose a point which is a fraction 0.38197 into the larger of the two intervals, from the central point of the triplet. The derivation can be found in Section 10.2 of Numerical Recipes.

We can now describe Brent's Method. If the function is nicely parabolic near the minimum then a parabola fitted through any three points ought to take us in a single leap to the minimum, or at least very near to it. Since we want to find an abscissa rather than an ordinate, the procedure is called inverse parabolic interpolation. The formula for the abscissa x that is the minimum of a parabola through three points $f(a)$, $f(b)$, and $f(c)$ is

$$x = b - \frac{1}{2}\frac{(b-a)^2[f(b)-f(c)] - (b-c)^2[f(b)-f(a)]}{(b-a)[f(b)-f(c)] - (b-c)[f(b)-f(a)]}$$

This formula fails only if the three points are collinear. Brent's Method works by using Golden Search when the function is not cooperative, but uses the formula above when the function allows. Brent's Method uses six variables, a, b, u, v, w, and x. The minimum is bracketed between a and b. x is the point with the very least function value found so far, or the most recent one in case of a tie. w is the point with the second least function value. v is the previous value of w. u is the point at which the function was evaluated most recently. There is an addition point x_m which is the midpoint between a and b, but the function is not evaluated here.

The Brent's Method code can be found in Section 10.3 of Numerical Recipes. Here is an overview of the algorithm.

1) Initialize the maximum number of allowed iterations. Initialize, a, b, x, w, v, and the values of f at w, v, and x to $f(x)$.

2) Begin the iteration. Calculated x_m. Test if the interval is less than the tolerance. If it is return the value of x and terminate.

3) Construct a parabolic fit. Check the acceptability of the fit to determine if it should use the Golden Search Method. Use this to find the step size for finding the next value to check. Use this value to update the variables. Repeat from step 2 until max iterations is reached or it terminates for the interval test.

18 Linear Conjugate Gradient

The conjugate gradient method is an iterative method for solving a linear system of equations

$$Ax = b$$

where A is an $n \times n$ symmetric positive definite matrix. The problem can be stated equivalently as the following minimization problem:

$$\min \phi(x) := \frac{1}{2} x^T A x - b^T x$$

This equivalence will allow us to interpret the conjugate gradient method either as an algorithm for solving linear systems or as a technique for minimizing convex quadratic functions, which is what we will be doing for the research project. For this problem, the gradient of ϕ equals the residual of the linear system,

$$\nabla \phi(x) = Ax - b := r(x)$$

A set of nonzero vectors is said to be conjugate with respect the symmetric positive definite matrix A if,

$$p_i^T A p_j = 0, \qquad \text{for all} i \neq j.$$

The importance of this set lies in the fact that we can minimize ϕ in at most n steps by successively minimizing it along the individual directions in a conjugate set. The conjugate gradient method has a very special property, namely that it can compute a new vector p_k by using on the previous vector p_{k-1}. It does not need to know all the previous elements of p. This property allows it minimize storage and computational cost. The search direction at iteration k is given by,

$$p_k = -r_k + \beta_k p_{k-1}$$

where β_k is a scalar "momentum" term that enforces the conjugacy of consecutive search directions. The step size is computed explicitly using a formula. We use the non-linear conjugate gradient method so further detail is not needed.

19 Non-Linear Conjugate Gradient

The Conjugate Gradient method is a minimization algorithm for the convex quadratic function ϕ. The question becomes, how do we adapt the approach to minimize general convex functions, or even general non-linear functions f. This is done with the Fletcher-Reeves and Polak-Ribiere Methods. The Fletcher-Reeves method makes two changes to extend the algorithm to non-linear functions. First the step length is calculated using a line search that identifies an approximate minimum of the non linear function f along p_k. This is the method we use for our research. We use Brent's method to find the value of the step size. Second we replace the residual r with the gradient of the non-linear objective f.

The Polak-Ribiere Method varies in the way it calculates β_k. The formula we use is in the section below.

20 Riemannian Conjugate Gradient Method

The Riemannian Conjugate Gradient Method is taken from the paper, Systematic Search for Singularities in 3D Euler Flows.

21 Spectral Methods

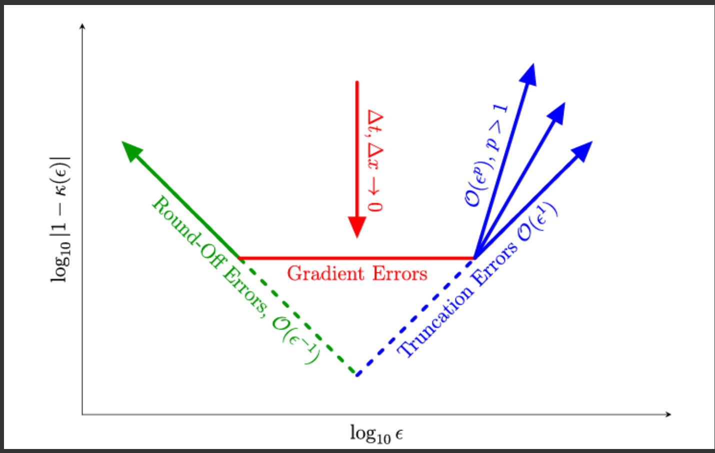
22 Optimization Problems

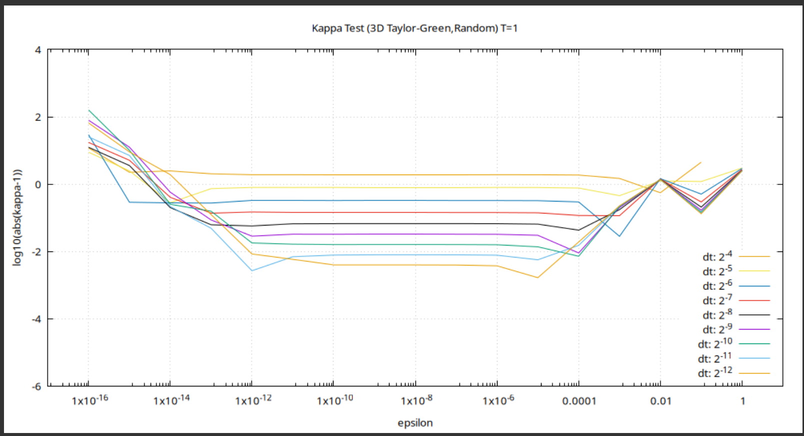
23 Kappa Test

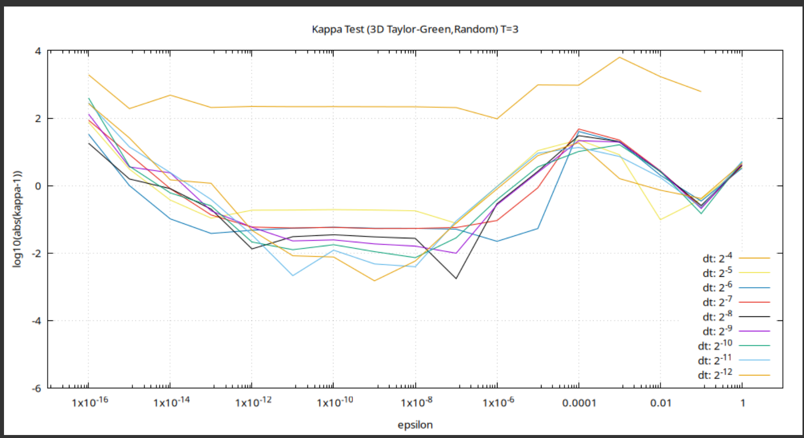
The Kappa test is a method for verifying the accuracy of the Gateaux differential. The formula for the kappa test is

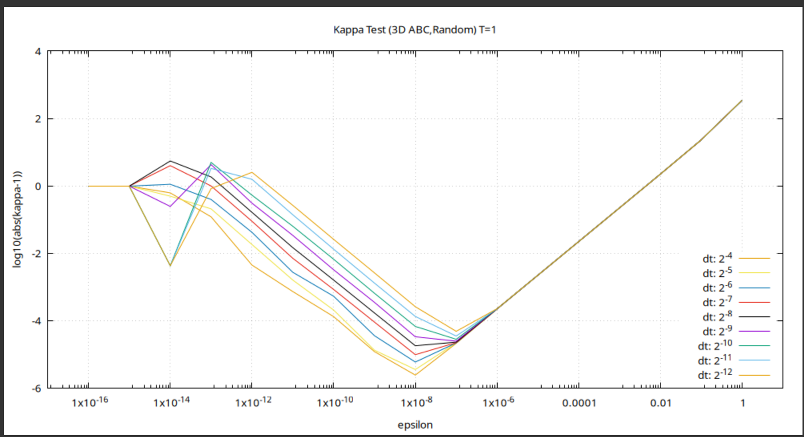
$$\kappa(\epsilon) = \frac{\epsilon^{-1}\left[\Phi(\eta + \epsilon\eta') - \Phi(\eta)\right]}{\langle \nabla\Phi(\eta), \eta' \rangle}.$$

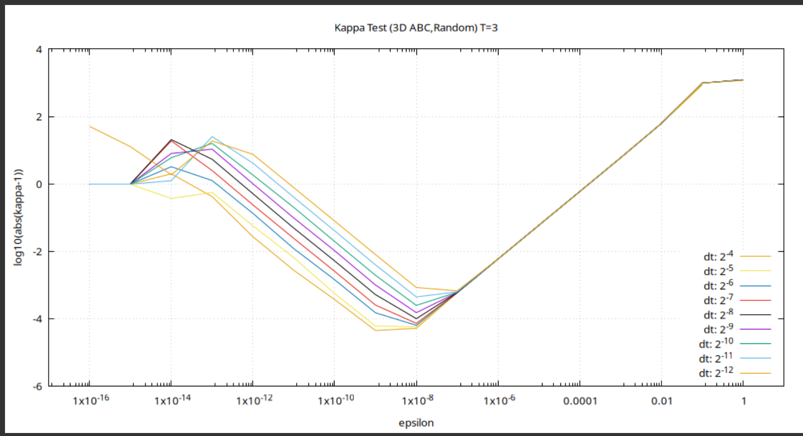
This formula shows the gradient, round off, and truncation errors as a function of epsilon. The round off errors should be approximately of order $O(\epsilon^{-}1)$. Therefore, we expect the value of κ do be dominated by round off errors for small ϵ values. The truncation errors should be of order $O(\epsilon^p)$ for some $p \geq 0$. Therefore, we expect the value of κ to be dominated by the truncation errors for large values of ϵ. Between these two regions the value of κ will be determined by the gradient errors, which are themselves a product of the spatial and temporal resolution. With smaller values of Δx and Δt causing the gradient error to decrease, while also decreasing the range of ϵ over which the gradient errors dominate. Therefore, we expect the plot of κ vs ϵ to be of the general form shown in the graph below, taken from Adjoint-based enforcement of state constraints in PDE optimization problems.
The kappa test graphs for our simulations are shown below.

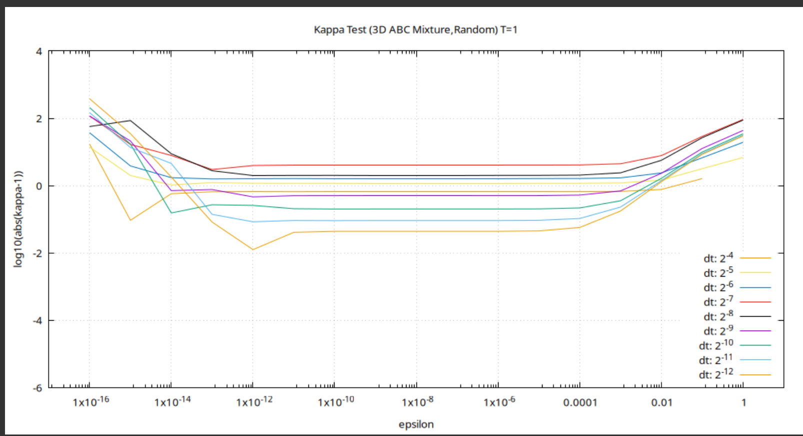

$\log_{10}|1-\kappa(\epsilon)|$
$\log_{10}\epsilon$
Round-Off Errors, $\mathcal{O}(\epsilon^{-1})$
Gradient Errors
$\Delta t, \Delta x \to 0$
Truncation Errors $\mathcal{O}(\epsilon^{1})$
$\mathcal{O}(\epsilon^{p}), p>1$

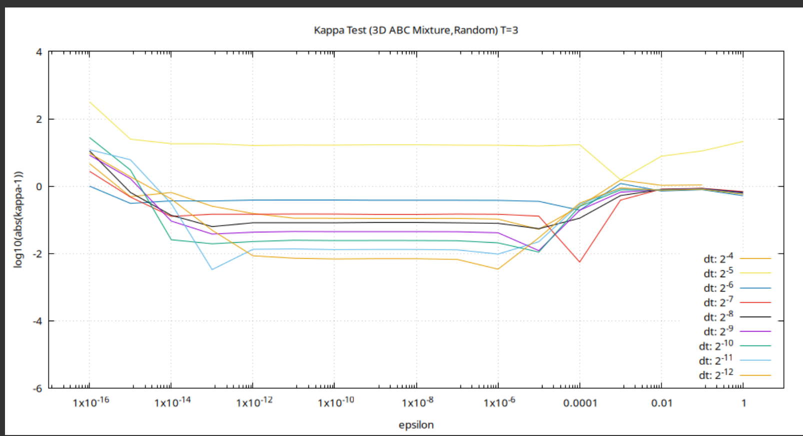

Kappa Test (3D Taylor-Green,Random) T=1
log10(abs(kappa-1))
epsilon
dt: 2-4
dt: 2-5
dt: 2-6
dt: 2-7
dt: 2-8
dt: 2-9
dt: 2-10
dt: 2-11
dt: 2-12


Kappa Test (3D Taylor-Green,Random) T=3
log10(abs(kappa-1))
epsilon
dt: 2-4
dt: 2-5
dt: 2-6
dt: 2-7
dt: 2-8
dt: 2-9
dt: 2-10
dt: 2-11
dt: 2-12


Kappa Test (3D ABC,Random) T=1
log10(abs(kappa-1))
epsilon
dt: 2-4
dt: 2-5
dt: 2-6
dt: 2-7
dt: 2-8
dt: 2-9
dt: 2-10
dt: 2-11
dt: 2-12







24 Paper Summary

A computational investigation of the finite-time blow-up of the 3D incompressible Euler equations based on the Voigt regularization

Adam Larios, Mark R. Petersen, Edress S. Titi, Beth Wingate

This paper examines two blow-up criteria for the 3D incompressible Euler-Voigt equations with periodic boundary conditions $\Omega := [0,1]^3$. It also investigates the analogous criteria for blow-up of the 1D Burgers equation, where blow-up is well known to occur. The result of the present work is that extrapolation to α suggests the development of a singularity in the 3D Euler equations corresponding to $T^* \approx 4.4 \pm 0.2$.

If the blow-up criteria are met, then the 3D Euler equations must develop a singularity at or before time T^*. The blow-up criteria are,

$$\sup_{t\in[0,T^*]} \limsup_{\alpha\to 0^+} (\alpha\|\nabla \mathbf{u}^\alpha(t)\|_{L^2}) > 0$$

and

$$\limsup_{\alpha\to 0^+} \left(\alpha \sup_{t\in[0,T^*]} \|\nabla \mathbf{u}^\alpha(t)\|_{L^2} \right) > 0$$

The second condition is a stronger criterion and so singularities indicated by the first will also be indicated by it.

All simulations were carried out using a pseudospectral method on the periodic unit cube, namely with derivatives computed in Fourier space, and products computed in physical space with the 2/3's dealiasing rule applied. Time stepping for the inviscid equations was done using a fully explicit fourth-order Runge–Kutta-4 scheme complying with the advective CFL condition. The Euler-Voigt simulations used Taylor-Green initial conditions with resolutions of 1024^3 and 512^3.

The paper also shows that the 1D Burgers equation developes a singularity at time $T^* = 1$ for initial condition $u_0(x) = -\sin(x)$.

25 Paper Summary

Maximum amplification of enstrophy in three-dimensional Navier-Stokes flows

Di Kang, Dongfang Yun, Bartosz Protas

This paper uses a systematic search for potential singularities in 3D Navier-Stokes flows. It uses Enstrophy as its criteria for identifying blow-up.

26 References